

HNSS17 · Hydrocephalus — Pathology Chart

Hydrocephalus: Types, Symptoms & Treatment

Definition

blocked CSF movement or failed absorption → ↑ CSF volume + pressure → enlarged ventricles

Obstructive

block inside the ventricles

Communicating

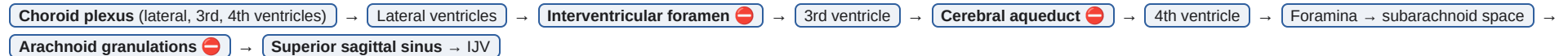
block between subarachnoid space and veins

NPH

elderly · wet, wacky, wobbly

TYPE	WHO / CAUSES	WHERE CSF IS STUCK	PATHOPHYSIOLOGY	SYMPTOMS & SIGNS	DIAGNOSIS & CLUES	TREATMENT & KEY FACTS
Obstructive (non-communicating) <i>the blocked hallway</i> INSIDE THE VENTRICLES	Congenital aqueductal stenosis , Dandy-Walker (4th ventricle cyst), Chiari II (STEP 1); tumors (e.g., medulloblastoma at the 4th ventricle), colloid cyst at the foramen of Monro (STEP 1)	Within the ventricular system: the interventricular foramen (of Monro) or the cerebral aqueduct (of Sylvius)	CSF keeps being made by the choroid plexus but can't leave the ventricles → ↑ volume + pressure upstream	Raised-pressure symptoms: headache, vomiting, sleepiness, confusion. Infants: enlarging head, bulging fontanelle (STEP 1)	Only the ventricles above the block enlarge (aqueduct block → lateral + 3rd ventricles big, 4th normal). LP is dangerous (herniation risk) (STEP 1)	Remove the blockage (e.g., tumor) or shunt ; supportive care
Communicating <i>the clogged drain</i> SUBARACHNOID SPACE → VEINS	Congenital absence of arachnoid villi ; villi blocked by RBCs after hemorrhage (e.g., SAH) or by protein after infection (meningitis)	Flow through the ventricles and subarachnoid space is NOT obstructed; CSF can't pass from the subarachnoid space into the venous system (superior sagittal sinus)	Production continues but absorption at the arachnoid granulations fails → CSF backs up everywhere	Raised-pressure symptoms (headache, vomiting, altered consciousness); history of hemorrhage or meningitis	All ventricles enlarge (open pathways). LP pressure ↑	Shunt ; treat the cause; supportive care
Normal pressure hydrocephalus (NPH) <i>wet, wacky and wobbly</i> ELDERLY	Usually elderly ; often idiopathic (can follow SAH, meningitis, trauma) (STEP 1)	Reduced CSF absorption (a communicating-type picture) (STEP 1)	Misnomer: CSF pressure is actually elevated across repeated measurements . Enlarged ventricles stretch the periventricular corona radiata fibers (STEP 1)	Diagnostic triad: urinary problems (wet), dementia (wacky), impaired gait (wobbly; broad-based, "magnetic") (STEP 1)	Ventricles enlarged out of proportion to sulci; gait improves after a large-volume tap → predicts shunt response (STEP 1)	Shunt (ventriculoperitoneal) — a reversible cause of dementia, unlike Alzheimer disease
Hydrocephalus ex vacuo <i>the look-alike</i> NOT TRUE HYDROCEPHALUS STEP 1	Brain atrophy: Alzheimer disease, advanced HIV, Pick disease	Nowhere — CSF flows and drains normally	Ventricles look big because brain tissue has been lost	Symptoms of the underlying dementia	Ventricles and sulci enlarged; normal pressure	No shunt; treat the underlying disease

Where CSF gets stuck



🚫 = where CSF gets stuck. A block at the foramen or aqueduct = **obstructive**; a block at the granulations = **communicating**. Normal: ~150 mL CSF, 400–500 mL made per day, pressure 70–180 mm H₂O.

★ **USMLE high-yield:** **Obstructive** = block **inside the ventricles** (interventricular foramen, cerebral aqueduct). **Communicating** = ventricles open, but CSF can't get from the subarachnoid space into the veins (absent villi; villi clogged by **RBCs after hemorrhage** or **protein after infection**). **NPH** = elderly, pressure actually elevated, **wet, wacky, wobbly**; shunt can reverse it. Treatment: **shunt**, remove the blockage, supportive care.

Comparing the Types & How Shunts Work

Obstructive vs communicating

	OBSTRUCTIVE (NON-COMMUNICATING)	COMMUNICATING
Where the block is	Between the ventricles and the subarachnoid space (foramen, aqueduct)	Between the subarachnoid space and the venous system (arachnoid villi)
Ventricles	Only those upstream of the block	All of them
CSF path through ventricles	Blocked	Open
Classic causes	Aqueductal stenosis, tumor	Hemorrhage (RBCs), infection (protein), absent villi

Treatment options

TREATMENT	WHEN
Shunting	Any type with ongoing CSF build-up; the main treatment for communicating hydrocephalus and NPH
Removal of blockage	Obstructive hydrocephalus from a removable cause (tumor, cyst)
Supportive care	While the cause is treated or the patient awaits surgery

Vignette decoder

CLUE IN THE STEM	THINK
Elderly + urinary incontinence + forgetfulness + shuffling, magnetic gait	NPH → shunt
Hydrocephalus a few weeks after subarachnoid hemorrhage or meningitis	Communicating (villi clogged by RBCs / protein)
Infant with a growing head; lateral + 3rd ventricles large, 4th normal	Obstructive at the cerebral aqueduct (aqueductal stenosis)
Child with a posterior fossa tumor, headache and morning vomiting	Obstructive at the 4th ventricle (medulloblastoma) STEP 1
Alzheimer patient with big ventricles and big sulci, normal pressure	Hydrocephalus ex vacuo (atrophy, not true hydrocephalus) STEP 1

★ **USMLE high-yield:** Remember the two block points: **ventricles → subarachnoid space** (obstructive) vs **subarachnoid space → venous system** (communicating). A **VP shunt** drains a ventricle to the peritoneum through a pressure-opening, adjustable valve (~1 h surgery, 50–80% success). NPH is a reversible dementia.

CSF shunts

PART / FACT	DETAIL
Inflow catheter	Sits in a ventricle and drains CSF
Valve	One-way; opens when pressure rises ; adjustable to CSF pressure
Outflow catheter	Runs through subcutaneous tubing , usually to the peritoneum (ventriculoperitoneal)
Surgery	About 1 hour
Success	50–80%
Complications STEP 1	Infection, obstruction, over-drainage

Signs of raised pressure that justify CSF analysis

Severe headache · stiff neck · confusion · seizures · persistent flu-like symptoms · fatigue · changes in consciousness · severe nausea · fever or rash · light sensitivity · poor coordination. **Rule out a mass / obstructive block before a lumbar puncture** (herniation risk) **STEP 1**